SE444: Introduction to Artificial Intelligence
Mid-Term II Fall 2025

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Date: Monday, November 10, 2025 Time: 5:30 PM – 7:00 PM (90 minutes)

Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Clearly state any assumptions made that might be needed to understand your solution.
- Show your work – any sign of serious effort will be considered.
- No marks will be given for not making any effort.

Student Information

Student ID:	Student Name:

Summary of Exam Questions

Part	Topic	Time	Score	Mark	Mapped CLOs
Part I MCQ	FOL — Empty Domain & Entailment	5 min	1		CLO 1
	Propositional Logic — Consistency	5 min	1		CLO 1
	CSP — Arc Consistency	5 min	1		CLO 1, 2
	Logical Agents — Architecture	5 min	1		CLO 1, 3
Part II FOL	Translation to FOL	10 min	3		CLO 1
	Translation with Negation	10 min	3		CLO 1
	Negation and CNF Conversion	20 min	4		CLO 1
Part III CSP	Formulate Lab Scheduling as CSP	10 min	2		CLO 2
	Apply Arc Consistency (AC-3)	15 min	3		CLO 2
	Find a Consistent Solution	5 min	1		CLO 2
	Total for Part I (MCQ)	20 min	4		
	Total for Part II (FOL)	40 min	10		
	Total for Part III (CSP)	30 min	6		
	Total Points	90 min	20		

Part I. Critical-Thinking Multiple Choice Questions

Total Score: 4 points **Total Time:** 20 minutes

Instructions: Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

Question I.1: First-Order Logic (FOL) — Quantifiers and Entailment

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

You are given the following two FOL sentences:

1. $\forall x\,(P(x) \rightarrow Q(x))$
2. $\exists x\,P(x)$

Which of the following sentences are logically entailed by (1) and (2)?

A) $\exists x\,Q(x)$

B) $\forall x\,Q(x)$

C) $\exists x\,(P(x) \wedge Q(x))$

D) $\forall x\,(Q(x) \rightarrow P(x))$

E) $\exists x\,(Q(x) \wedge \neg P(x))$

Question I.2: Propositional Logic — Consistency vs. Entailment

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

Consider a knowledge base (KB) with propositional atoms p, q, r, s, t:

1. $(p \wedge q) \rightarrow r$
2. $r \rightarrow (s \vee t)$
3. $\neg s \wedge \neg t$
4. q
5. $p \vee \neg r$

Which of the following statements is true?

A) KB is unsatisfiable.

B) KB $\models p$.

C) KB $\models \neg p$.

D) KB $\models (r \wedge s)$.

E) KB can be satisfied with $p = \text{true}$ and $r = \text{true}$.

Question I.3: CSP — Arc Consistency (AC)

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 2

Consider a CSP with variables A, B, C.
Initial Domains: $D(A) = \{1, 2\}$, $D(B) = \{1\}$, $D(C) = \{1, 2\}$
Constraints:

- $A \neq B$
- $B < C$
- $C \leq 2$ (already reflected in $D(C)$)

Run AC-3 (Arc Consistency) checks and remove impossible values until nothing changes.

Which are true?

A) After AC, $D(C)$ becomes $\{2\}$.

B) After AC, $D(A)$ becomes $\{2\}$.

C) After AC, all domains become singletons and the CSP is solved without search.

D) If we replace "$B < C$" by "$B \leq C$", AC would still make all domains singletons.

E) If we keep "$B < C$" but widen $D(B)$ to $\{1, 2\}$, AC must still prune C to $\{2\}$.

Question I.4: Logical (Knowledge-Based) Agents — Architecture vs. Environment

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 3

A company deploys **autonomous delivery drones** operating in a complex city environment that is **partially observable** (limited sensors and occasional signal loss), **stochastic** (uncertain wind and traffic conditions), **sequential** (each action affects future deliveries), and **dynamic** (the world keeps changing while the agent is acting).

Evaluate the following statements about possible agent architectures and identify which are correct.

A) A simple reflex agent with a fixed condition–action table can act rationally in such an environment if its rule base is sufficiently large to cover all possible situations the drone may face during operation.

B) A model-based reflex agent can outperform a simple reflex agent in this scenario by maintaining an internal belief about unobserved factors such as wind direction, obstacle positions, or battery status, allowing it to make more informed decisions.

C) A goal-based agent can only behave rationally if the effects of all actions are deterministic and known in advance, since uncertainty in outcomes would prevent it from reliably selecting actions that achieve its goals.

D) A utility-based agent can prefer actions that reduce uncertainty — such as temporarily rising to a higher altitude to gain better GPS visibility — even when these actions do not immediately improve goal satisfaction.

E) In such environments, table-driven agents are always optimal if the table covers every possible percept sequence, regardless of changes or randomness in the environment.

Part II. Logical Reasoning in Education Context

Total Score: 10 points **Total Time:** 40 minutes

Given Predicates:

Predicate	Meaning
Student(x)	x is a student
Teacher(x)	x is a teacher
Course(x)	x is a course
Takes(x, y)	Student x takes course y
Teaches(x, y)	Teacher x teaches course y
Understands(x, y)	Student x understands topic y
Topic(y)	y is a topic
Covers(c, y)	Course c covers topic y
GradedBy(x, t)	Student x is graded by Teacher t
Passes(x, y)	Student x passes course y

Translate each of the following statements into first-order logic (FOL) using the provided predicates, then perform the required logical transformation.

Question II.1: Translation to FOL

Time: 10 minutes **Score:** 3 points **Mapped CLOs:** CLO 1

Translate the following statement into first-order logic (FOL):

"Every student who takes a course taught by a teacher understands at least one topic that the course covers."

Question II.2: Translation with Negation

Time: 10 minutes **Score:** 3 points **Mapped CLOs:** CLO 1

Translate the following statement into first-order logic (FOL):

"No teacher grades a student who is not enrolled in one of their courses."

Question II.3: Negation and Conjunctive Normal Form (CNF)

Time: 20 minutes **Score:** 4 points **Mapped CLOs:** CLO 1

Negate the following statement and convert it step-by-step into Conjunctive Normal Form (CNF). Show all transformations clearly and justify each step using the appropriate logical rule (e.g., $\neg\forall$, implication removal, De Morgan's law, or Skolemization).

$$\forall x\,(\text{Student}(x) \rightarrow \text{Hardworking}(x))$$

Part III. Constraint Satisfaction Problem (CSP)

Total Score: 6 points **Total Time:** 30 minutes

Part III: Constraint Satisfaction Problems (CSP)

Context: You are designing the schedule for **AI Lab presentation sessions** on Mondays. There are three labs and three teaching assistants (TAs) available. Each TA can supervise at most one lab at a time, and some labs overlap in time. Your goal is to assign each lab to a TA in a way that satisfies all constraints.

Labs and Meeting Times

Lab	Topic	Time
Lab 1	Search Algorithms	8:00–9:00 am
Lab 2	Machine Learning	8:30–9:30 am
Lab 3	Knowledge Representation	9:00–10:00 am

TA Availability

TA	Available for Labs
TA A	Labs 1 and 2
TA B	Labs 2 and 3
TA C	Labs 1, 2, and 3

Assignment Rules (Constraints)

1. Each lab must be supervised by exactly one TA.
2. A TA cannot supervise overlapping labs (time conflicts).
3. A TA can only be assigned to labs where they are available.

Question III.1: Formulate as a CSP

Time: 10 minutes **Score:** 2 points **Mapped CLOs:** CLO 2

You are asked to formulate the problem of the scheduling scenario as a **Constraint Satisfaction Problem (CSP)**.

Clearly specify:

- the **variables**,
- the **domains** of each variable, and
- the **binary constraints** that must hold between the variables.

You may describe the constraints in English words.

Question III.2: Apply Arc Consistency (AC-3)

Time: 15 minutes **Score:** 3 points **Mapped CLOs:** CLO 2

Using the CSP formulation from Question III.1, apply the **Arc Consistency (AC-3)** algorithm.

- Check each relevant arc between lab variables based on the binary constraints.
- For each value in a variable's domain, determine whether it has at least one supporting value in the connected variable's domain.
- Indicate which (if any) domain values are removed and explain why.
- Continue until a **fixed point** is reached (no further pruning possible).
- State the **final domains** for all variables after enforcing arc consistency.

Hint: For constraints of the form $L_i \neq L_j$, a value x in $D(L_i)$ is supported if there exists at least one y in $D(L_j)$ such that $x \neq y$.

Question III.3: Find a Consistent Solution

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 2

Based on your AC-3 results from Question III.2, provide **one valid assignment** that satisfies all constraints and briefly explain why this assignment satisfies all constraints.